

Waitlist COLA: A Record-Blind NBA Draft Order With No Tanking Reward

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

Introduction

A team eliminated from playoff contention improves its next draft position by losing, and roughly a third of the National Basketball Association’s teams were openly doing so in the 2025-26 season. For a decade the league answered by re-weighting its lottery odds, and in May 2026 adopted a wider sixteen-team format called 3-2-1. Banchio and Munro (SSAC 2020) proved that any rule giving worse end-of-season records better draft odds must reward losing.

Carry-Over Lottery Allocation (COLA) escapes that impossibility by changing what the draft reads. Every team carries an index that grows each season it misses the playoffs and falls when it succeeds, so priority tracks how long a franchise has waited and one season moves it little. COLA is a family whose members differ on who accumulates the index, what success spends, and how the index becomes picks, and none has been measured against the others. We measure them, and the answer turns on a part of the lottery no reform has touched.

Methods

We formalize the family as seven design dials and evaluate nine mechanisms across 14,004 league-seasons of ZenGM Basketball-GM, an open-source simulator that plays every game with generated rosters and automated general managers. Each runs 48 independent leagues of 15 seasons, with uncertainty at the replicate level. Each is scored on one statistic read twice, the gap in draft position between the first- and fifth-ranked team, ranked by this season’s record and by the mechanism’s own priority metric. A paired counterfactual then moves one team’s record with the season held fixed.

What the rule reads	Reward for losing	Share
Both components read record (rule through 2026)	1.30	100%
Draw made record-blind, queue by record	1.98	152%
Queue made record-blind, draw unchanged	0.61	47%
Neither component reads record	−0.16	noise floor

Table 1. Where the reward for losing lives. Through the 2026 draft four picks were drawn and picks 5 through 14 followed reverse record, and each row switches one component to a record-blind rule. Values are places of draft position gained by the worst-record team over the fifth-worst.

Results

The reward for losing sits mostly in the ten picks awarded after the televised draw, and making the draw record-blind raises it. The adopted format confirms this. Holding a 3-2-1-shaped ball allocation fixed and drawing only four picks leaves +1.76, worse than the rule it replaced; drawing all sixteen, as adopted, takes it to −1.22. Its whole gain against tanking comes from how deep it draws.

Wherever a rule treats the worst record as most deserving, the two readings in Table 2 are one number, the impossibility made visible. One configuration separates them, and we name it *Waitlist COLA*. Dropping a team from fifth-worst to worst record buys 1.87 places under Classic COLA and 1.32 under the 2026 rule. Under Waitlist COLA it buys exactly zero across 720 seasons. The pattern repeats on 26 seasons of actual NBA results.

Mechanism	Ranked by record <i>want 0</i>	Ranked by own metric <i>want wide</i>
NBA lottery, 2019 rules	+1.09 ± 0.12	+1.09 ± 0.12
3-2-1 as adopted	−1.22 ± 0.25	−1.22 ± 0.25
Classic COLA (original)	+1.88 ± 0.11	+1.38 ± 0.24
Waitlist COLA	−0.12 ± 0.22	+1.71 ± 0.13

Table 2. One statistic under two rankings. Waitlist COLA draws four picks by index, orders the rest by index, and breaks ties at random. Record-based rules post identical columns by construction, so they cannot aim help without paying for losing.

Conclusion

Waitlist COLA changes one thing about the original variant, and only the picks awarded after the televised draw. The format the league adopted removes the reward by pushing its worst records down the order. Waitlist COLA removes it while drafting the longest-waiting teams earliest, and the seven dials cover the rest of the design space. The adopted format expires after the 2029 draft, when a fresh vote reopens the question.